The American Pancake & Waffle — One Soak, Two Machines

One overnight soak of whole grain and cultured dairy. Griddle it for a tall, mahogany pancake — or change three numbers and send it to the iron. The split happens twenty minutes before service, not the night before.

Yield: **36 pancakes** or **28 waffles** (10–12) Active: **35 min** Total: **overnight** Griddle: **175°C** · Iron: **highest**

WHERE IT WAS BORN
North America — the tall griddle cake follows the leavener. *Pearlash* (potassium carbonate, leached from wood ash) went into print in Amelia Simmons' *American Cookery*, Hartford 1796; *saleratus* by the 1840s; Horsford's calcium acid phosphate patented 1856.

EATEN, AUTHENTICALLY
Stacked and dealt from a shared pile at a slow breakfast — butter melting between the layers, the syrup passed down the table rather than poured for you.

HOW IT'S USED NOW
The diner stack, everywhere. This build keeps the height and the browning and runs the flour **100% whole grain** as a pancake (~91% as a waffle) — about 15 g protein and 5.5 g fibre a serving.

INGREDIENTS

WEIGHT (LB/OZ) · CUPS / VOLUME IN RIGHT COLUMN

THE SOAK — NIGHT BEFORE		
Cottage cheese *(4%, blended smooth)*	14.1oz	1¾ cups
Buttermilk	1lb 7.3oz	2⅔ cups
Oat flour *(or oats, blitzed)*	10.6oz	3⅛ cups
White whole wheat flour	10.6oz	2⅜ cups
Fine sea salt	0.32oz	1½ tsp
THE WAKE — 20 MIN BEFORE SERVICE		
Brown butter *(from 100g butter)*	3.2oz	⅜ cup
Egg yolks	6	keep whites
Maple syrup *(in it, not on it)*	1.4oz	2 tbsp
Baking powder	0.53oz	1⅛ tbsp
Bicarbonate of soda	0.18oz	1⅛ tsp
THE FOLD		
Egg whites, cold	6	from above
Butter, for the griddle	–	

THE RATIO — FLOUR AT 100%
Flour 100% (½ oat · ½ white whole wheat) · Buttermilk 110% · Cottage cheese 67% · Egg 50% · Brown butter 15% · Maple 7% · Salt 1.5% · Powder 2.5% · Soda 0.8%. Dairy at **177%** against a diner pancake's 120% — whole grain holds far more water, so build it wet enough to still pour *after* it drinks. Soda capped at 0.8%: past what the acid can spend, you taste soap.

Kit: blender · scale · two large bowls · whisk · balloon whisk or mixer · flat griddle or wide non-stick · 60g ladle · wire rack · lidded container for the soak

METHOD

1. **Soak — night before.** Blend 14.1oz cottage cheese with 1lb 7.3oz buttermilk until **completely smooth** — run it longer than feels necessary. Whisk in both flours + 0.32oz salt. Cover, fridge overnight. It stiffens to wet plaster. That is right.
   > *Bran and oat β-glucan drink slowly. Water they haven't taken by the pan, they take from the crumb as it sets — that is the gummy middle. Acid also wakes the grain's own phytase: at pH 5.5 phytate breakdown runs 40%→70%, freeing iron and zinc.* — Leenhardt et al. 2005
2. **Warm soak for the minerals.** Cold overnight = texture only. **2–4 hr at room temperature** is what actually moves the phytate. Hot galley: fridge, and take the texture win.
3. **Brown the butter.** 3.5oz butter, pale pan, medium. Foam falls, solids go hazelnut. Off **the moment it smells of nuts** — colour lags flavour ~20 sec. Scrape the solids in. Cool to warm.
   > *Milk protein + lactose running Maillard in hot fat. At ~7 g of butter a head the fat budget is lean, so it has to be loud.*
4. **Wake it.** Stir in 6 yolks, the warm brown butter, 1.4oz maple. **Sift** 0.53oz powder + 0.18oz soda over — tipped in dry it leaves metallic pockets. Loosen with buttermilk until a ladleful **ribbons and sinks back in 3 sec.**
   > *Soda meets the buttermilk acid at once (~1.4 L of gas). Spending that acid walks the batter from pH 4.5 toward neutral — step 6 collects on it. The powder is double-acting: the reserve that keeps the last ladle as tall as the first.*
5. **Whip & fold.** 6 whites to **soft peak** — tip flops, doesn't stand. Fold in three, stop while streaky. **Cook within 25 min.**
   > *Half this flour is oat — no gluten at all. Whipped whites go around the problem: egg proteins unfold at the air-water interface and set a film that holds bubbles by itself. Past soft peak the foam shatters on folding instead of stretching.*
6. **Griddle low and long. 175°C** (350°F) — lower than a white-flour pancake. Butter on, wiped almost dry. **60g** a cake, spread to 10cm. **2½–3 min**, flip **once**, **1½–2 min**.
   > *Maillard needs an unprotonated amino group; acid holds it protonated, alkali frees it, so browning climbs steeply with pH. That is how a batter with 3 g of sugar a serving still lands mahogany — the colour is bought with pH, not sugar. It also browns early, so the heat comes down to let the middle finish.* — Ajandouz & Puigserver 1999
7. **Rack it, single layer.** Wire rack, **uncovered**, 70°C oven. Cover them and the crust goes to flannel in five minutes.

Done: the bubbles opening at the rim **stay open** — a batter that reseals its own holes is still too wet to turn. Rim matte from the outside in, ~5mm deep. Underside even mahogany, not blotched. Springs back slowly at the centre with no wet give underneath.

Service — the split batter is the whole trick: the soak holds **48 hr**, the folded batter holds **nothing**. Staggered service? Hold back a third of the soak and fold fresh whites into it later — never re-whip a slumped batter, and never leaven the whole soak unless you are cooking the whole soak.

ONE SOAK, TWO MACHINES — THE WAFFLE
Soak unchanged. At step 4, three numbers move: **brown butter 3.2oz→7.4oz** (fat fries the plate face and releases from the iron) · **+2.1oz cornflour** sifted in (pure starch, no soak needed; goes glassy where β-glucan can't — takes it to ~91% whole grain) · **+5.3oz buttermilk** (it must *flow* into the grid, not sit on it). Iron on **highest**, preheat 10 min, **5–7 min** — longer, never hotter; the shell can't brown until it has dried. **Done:** hard steam drops to a thin wisp; the lid lifts with no resistance. **Rack, never stack.**
A pancake is a sponge, a waffle is a plate — the pancake drinks butter and syrup; the waffle's pockets hold wet things on top (poached egg, creamed mushrooms). **Pancakes = crew breakfast** (live, eaten in 10 min). **Waffles = charter service** (cook ahead, freeze flat, toast from frozen — they come back crisper).

ELEVATION

TIER 1 — NO EXTRA TIME
- **Frozen blueberries** — onto the wet face, never stirred in
- **Flaky salt** — on the stack, not in it
- **Lemon zest** — into the soak

TIER 2 — WORTH IT
- **Buckwheat** — for 100g of the oat; brown 10°C cooler
- **Barley flour** — same β-glucan; +5% buttermilk
- **Kefir** for buttermilk — soda down to 4g

TIER 3 — SERVICE
- **Waffle** + poached egg — the grid holds the yolk
- **Pancakes** stacked — roast stone fruit, thick yogurt
- **Crew** — 120g pancake folded over bacon

TROUBLESHOOT

Gummy, damp middle	Reached the pan still thirsty, or griddle too hot → longer soak; 165°C + a minute
Soapy / metallic	More soda than the acid can spend, or unsifted → drop to 4g, or +60g buttermilk
Flat, dense base band	Whites stiff and broken, or batter sat → soft peak; cook within 25 min
Pale and dull	Griddle cool, or soda under-dosed → fix pH before reaching for sugar
Bitter, bran-y	Red wholemeal used → hard *white* wheat; same grain, no tannins
Rubbery flecks	Cottage cheese curd survived → blend longer or sieve it

CHARTER PREP & STORAGE

Best route	Soak the night before; soda and whites at service only
Soak	48 hr fridge; improves over the first 24
Folded batter	25 min, then it slumps — no holding, ever
Hold	Wire rack, single layer, uncovered, 70°C, up to 40 min
Frozen	Waffles freeze *better* than fresh — toast from frozen. Pancakes: 180°C 4 min, never quite as good
Provisioning	*White* whole wheat, not red — the one call guests notice

CONTACT THE CHEF @arnaudcallier www.littoralicious.com

Science: Leenhardt et al. 2005 (*J. Agric. Food Chem.*) · Ajandouz & Puigserver 1999 · Aune et al. 2016 (*BMJ*) · Lo & Lampoglou et al. 2026 (*npj Sci. Food*) · EFSA 2010